PennState

Department of Human Development and Family Studies
College of Health and Human Development
The Pennsylvania State University
119 Health and Human Development Building
221 E. College Avenue
University Park, PA 16802

814-863-8000
fax: 814-863-7963
HDFSinfo@psu.edu

October 23, 2026

Emilio Ferrer, Ph.D.
Editor-in-Chief, *Multivariate Behavioral Research*

Dear Dr. Ferrer:

We are pleased to resubmit our revised manuscript, "From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models" (MBR-26-107), for consideration in Multivariate Behavioral Research.

Thank you and the Associate Editor and reviewers for the thoughtful assessment of our work. The comments helped us clarify the connection between idiographic modeling and meta-analytic SEM and strengthen the evaluation of MetaVAR. The revision examines zero and larger transition heterogeneity, compares DSEM under two prior specifications in the simulation and four in the empirical illustration,~~compares two DSEM prior specifications,~~ distinguishes sampling uncertainty from between-person variation, and adds an empirical comparison with the uncertainty-uncorrected approach.

We have also clarified the limits of the evidence. The paper discusses higher-dimensional estimation as a limitation, reports variance-boundary difficulties, and reports what the empirical DSEM fits under weak and data-informed priors can and cannot show.~~explains why the empirical DSEM fits do not support a reliable numerical comparison.~~ The revised interpretation emphasizes MetaVAR's contribution to propagating uncertainty in person-specific estimates while acknowledging conditions in which DSEM performs better or simpler summaries give similar results.

The point-by-point responses below reproduce the reviewers' comments and identify the changes and their locations. Page references refer to the supplied manuscript.pdf and supplement.pdf, each numbered separately; they do not refer to the tracked manuscript. Reviewer 2's unnumbered comments are grouped into 16 items for ease of reference.

Thank you for considering the revised manuscript.

Sincerely,

Ivan Jacob Agaloos Pesigan, Ph.D.
Assistant Research Professor
Quantitative Developmental Systems Methodology Core (QuantDev)
Email: ijapesigan@psu.edu

Response to the Associate Editor and Reviewers

Associate Editor

Comment AE-1

> 1) Both reviewers mentioned the magnitude of heterogeneity in parameters, but they had opposite recommendations (for different reasons). Reviewer 1 suggested using larger variances (to offset the impact of priors) and Reviewer 2 suggested smaller variances (to assess how noisy estimate would be; although reviewer two also mentioned that low but non-zero population values did make some simulation results less clear, similar to Reviewer 1). I think both points are compelling; my takeaway was that some additional justification for the population values might be helpful (possibly including additional text in the discussion to address how results may potentially change for conditions that were not studied).

Response. ~~Thank you for bringing these complementary concerns together.~~ The revised simulation now includes zero, reference, and doubled transition heterogeneity, with both default and alternative DSEM priors. We explain why the reference values were chosen and distinguish small nonzero variation from exact homogeneity. The expanded Results show how the methods respond to these conditions: MetaVAR's transition-variance coverage advantage is clearest at reference heterogeneity, while DSEM becomes more competitive as heterogeneity increases. We also acknowledge that varying heterogeneity around one mean dynamic system does not establish performance across all generating processes.

Location in the revision. Manuscript, pp. 14, 21, 32. Heterogeneity design, simulation findings, and scope of generalization.

Comment AE-2

> 2) Both reviewers also mentioned that performance with a larger number of variables may be of interest given the potential complications that may arise under these conditions. If this is outside of the paper's scope, the discussion should incorporate some text to outline how performance may possibly be affected by a larger number of variables.

Response. We have taken the discussion-based approach you suggested. The General Discussion now explains how the number of features and between-person covariance components grows with the number of variables, why both individual series length and participant sample size matter, and how covariance restrictions might help. We explicitly limit the performance conclusions to the bivariate evaluation. The computational advantage observed in the benchmark is not presented as evidence that an unrestricted higher-dimensional model will be estimable.

Location in the revision. Manuscript, pp. 32–33. Dimensionality and covariance restrictions in the General Discussion.

Comment AE-3

> 3) There are several comments suggesting potential clarifications. I think the idea to link these two frameworks is insightful, but a potential challenge of blending two different frameworks is that

readers with expertise in MASEM may not necessarily be familiar with idiographic modeling and vice versa. As a global comment, during the revision, I would just suggest that the authors pay special attention to making sure that the paper is accessible to the broader audience that may be interested in this work, which may include readers who only have knowledge of one of methods being combined.

Response. We revised the exposition where the connection between the two frameworks needed to be clearer. ~~Your accessibility concern helped us identify where the connection between the two frameworks needed to be clearer.~~ The Stage-1 exposition now avoids or explicitly defines jargon that may have very different meanings in the MASEM and VAR frameworks (e.g., "heterogeneity parameters" as brought up by R2 as a point of confusion), and clearly distinguishes a person's estimated dynamic parameters, their sampling covariance, and the innovation covariance within that person's VAR. The Stage-2 exposition then explains how person-specific ~~howperson-specific~~ estimates with uncertainty support inference about population means and between-person variation. We have also clarified the simulation targets before the Results and organized the findings around patterns, with numerical detail in the figures. These changes preserve the paper's existing progression.

Location in the revision. Manuscript, pp. 7–8, 18, 20. Stage 1 exposition, performance targets, and Results.

Reviewer 1

Comment R1-1

> 1. The simulation compares MetaVAR against a Bayesian multilevel VAR (BMLVAR) implemented in Mplus using default or "relatively uninformative" priors. The authors themselves note that "recovery of multilevel variance–covariance components can depend materially on prior specification, especially when heterogeneity parameters are small, weakly identified, or near the boundary" (p. 16). The DGP in this simulation has small heterogeneity variances for the transition parameters: the diagonal entries of Σ_A are 0.017, 0.008, 0.001, and 0.026 (Equation 11). These are exactly the conditions known to introduce substantial bias and undercoverage. This does not necessarily indicate MetaVAR is better than BMLVAR. The authors may need to consider other common priors in the BMLVAR to address this concern.

Response. We agree that the original comparison could not separate a method difference from sensitivity to priors near small variances. The revision therefore reports default and alternative DSEM priors throughout, together with doubled transition heterogeneity. The Methods give the alternative hyperparameters and explain their rationale; the supplement notes that the proper variance priors remain heavy-tailed and influential near zero.

The interpretation has changed accordingly. DSEM generally recovered transition means better, and its variance coverage became more competitive at larger heterogeneity. Alternative priors also improved covariance point recovery in many conditions. MetaVAR's contribution is framed around propagation of first-stage uncertainty and its demonstrated heterogeneity-recovery benefits, rather than superiority to Bayesian estimation in general. We also examined convergence more closely than the default `Mplus` check, which is based on the potential scale reduction factor. For every parameter in every replication, we computed the rank-normalized $\widehat{R}$, bulk and tail effective sample sizes, and Monte Carlo error, and Supplement Table 1 reports how many replications met all of these criteria. The primary results retain every replication, as a typical `Mplus` analysis would.~~We also report posterior-quality counts so that normal termination is not mistaken for adequate mixing.~~

We distinguish available DSEM summaries from satisfactory posterior sampling. Every required parameter had to satisfy the convergence, effective-sample-size, and Monte Carlo error criteria for a replication to pass. The primary results retain all usable runs, including those failing this screen. Consequently, the DSEM comparisons describe the evaluated priors and computational settings, rather than performance guaranteed to reflect fully converged posterior sampling. Because the strict thresholds are demanding, we also report a moderate criterion ($\widehat{R} \leq 1.05$, effective sample sizes of at least 100, relative Monte Carlo error at most .10). Most strict failures came from the random-effect parameters and fell between the two sets of thresholds: averaged over conditions, random-effect failure rates dropped from .697 to .013 under default priors and from .909 to .235 under alternative priors, with the remaining alternative-prior failures concentrated in the 50-occasion conditions. Supplement Table 2 gives these rates for every condition.

We did not add a DSEM branch with priors centered on the generating values. Such priors would give DSEM population information that an analyst does not have and that MetaVAR does not use, so the comparison would partly measure prior information rather than estimator performance. Instead, the empirical illustration now includes DSEM fits with data-informed priors centered on the MetaVAR estimates, the data-based specification that *Schuurman, Grasman, and Hamaker (2016)* recommend including in prior sensitivity analyses. These fits converged where the weak-prior fits did not, but their transition covariances followed the prior's scale matrix,

which illustrates your point that small heterogeneity components are sensitive to the prior. The Discussion now notes that data-based and half-t priors were not evaluated in the simulation and could change the DSEM results.

The revised Methods (DSEM) read: "Neither branch centers the priors on the generating values. Such priors would give DSEM population information that is unavailable in practice and that MetaVAR does not use, so the comparison would partly measure prior information rather than estimator performance." They also add: "Because these thresholds are demanding for multilevel models with many random-effect parameters, we also applied a moderate criterion as a sensitivity analysis: $\widehat{R} \leq 1.05$, both effective sample sizes at least 100, and relative Monte Carlo error at most 0.10." The General Discussion adds: "The simulation also evaluated DSEM under only two prior specifications. Data-based inverse-Wishart priors, which performed best in a comparison of inverse-Wishart specifications for multilevel autoregressive models (Schuurman, Grasman, & Hamaker, 2016), and half-t priors on the random-effect standard deviations (Gelman, 2006) were not evaluated."

Location in the revision. Manuscript, pp. 4, 17, 20–21, 30; Supplement, pp. 2, 6. DSEM priors and diagnostic criteria, usable-run reporting, and interpretation; supplement prior note and Table 1; the moderate criterion in the Methods and Results and in supplement Table 2; the empirical DSEM fits with data-informed priors (Practical and Methodological Considerations); the prior limitation in the General Discussion.

Comment R1-2

> 2. The entire simulation is conducted with k = 2 variables. For a bivariate VAR(1), the Stage 1 parameter vector θ_i has dimension p = 9 (two set points, four transition coefficients, three innovation parameters). The between-person covariance τ^2 is a 9 × 9 matrix, estimated from N ∈ {50, 100, 200} persons. In real EMA applications, k ≥ 4 variables are common (e.g., distinct positive and negative affect facets, stress, arousal). It is unclear whether MetaVAR is practically estimable in higher-dimensional settings, and no guidance is provided. A related question is that only one set of population values (Eq. 11) was used, making it unclear how generalizable the findings are to other settings.

Response. We agree that the bivariate evaluation leaves an important question about practical estimability. We kept the bivariate design because it isolates the question the simulation was built to answer: whether carrying Stage 1 sampling covariance into Stage 2 changes inference about population means and between-person heterogeneity, relative to a one-step estimator and an uncertainty-uncorrected two-stage workflow. Holding the number of variables fixed allowed us to vary participant sample size, series length, unequal series lengths, the magnitude of transition heterogeneity (including zero), and the DSEM prior within one design. The results therefore show where uncertainty propagation helps and where it does not in a system small enough for every method to be fitted across all 36 conditions. They do not show how MetaVAR performs with more variables, and we now treat that question as a future direction.~~We first clarified the dimensions actually fitted: Stage 1 estimated a nine person-specific parameters (features), but the simulation synthesizes six and uses separate set-point and transition blocks, with 13 heterogeneity components. The empirical model uses all nine features and 45 components.~~

The General Discussion now addresses the demands of larger systems and the potential role of covariance restrictions, following the Associate Editor's suggestion to discuss this limitation. Zero and doubled heterogeneity broaden the simulation, but we explicitly acknowledge the remaining restrictions to one mean system, fixed innovation covariance, and correct specification. We do not claim demonstrated higher-dimensional performance.

Location in the revision. Manuscript, pp. 17, 32–33. Six-feature simulation specification and dimensionality limitations.

Comment R1-3

> 3. The between-person covariance τ^2 is a p × p matrix estimated from N persons. Near the boundary of the positive-semidefinite cone, standard ML confidence intervals lose their asymptotic validity. The paper does not report the rate of boundary solutions in the simulation, how near-zero diagonal elements are handled (are they fixed to zero? constrained?), or what the implications of near-boundary estimates are for reported coverage rates. Given that several diagonal elements of τ^2 under the DGP are small (e.g., the off-diagonal transition parameters have very small population variances), boundary estimation is plausible for smaller N and T cells.

Response. This was a consequential omission. The supplement now specifies the covariance parameterization, defines near-zero variances and near-singular matrices, and reports their frequencies by condition alongside usable replication counts. Near-boundary estimates are retained when otherwise usable; they are not silently discarded or fixed to zero.

The Results now make the inferential limitation explicit. At zero transition heterogeneity, normal-theory interval exclusion ranged from .004 to .398, and robust intervals did not resolve the problem. We distinguish near singularity from a near-zero diagonal and explain why Bayesian credible intervals do not perform satisfactorily as an ~~an~~ inferential test for variance-related parameters that are close to 0. The Discussion also separates assuming a common effect from estimating heterogeneity at the boundary.

We added two analyses aimed at inference near the boundary. First, because a variance can be negligible without being exactly zero, we evaluated practical homogeneity with a tolerance, following the logic of equivalence testing. With a tolerance of .0009 (a between-person standard deviation of .03, the benchmark for a small cross-lagged effect), MetaVAR's lower limits exceeded the tolerance for .023 of the transition-variance estimates at zero heterogeneity, against 1.000 for Uncorr and .761 for default-prior DSEM; Supplement Table 3 reports two further tolerances. Second, we implemented a boundary likelihood ratio test for the transition covariance block, referred to a chi-bar-square distribution with simulated mixing weights rather than the usual chi-square. In the empirical illustration, the test rejected homogeneous transitions (statistic 17,342 on 30 constraints, $p < .001$).

Add the simulation results of the boundary LRT (power and Type I error by condition) and the Stage 1 calibration explanation after the summary rerun.

The revised Results (Heterogeneity Parameters) read: "Testing an exact zero asks more than most applications need, because a variance can be negligible without being zero. We therefore also evaluated practical homogeneity of the four transition variances against a tolerance δ, following the logic of equivalence testing (Lakens, Scheel, & Isager, 2018)." The empirical Results add: "Because the null hypothesis places variances on the boundary of the parameter space, the statistic does not follow the usual χ^2 distribution."

Location in the revision. Manuscript, pp. 21–23; Supplement, pp. 2, 6. Zero-heterogeneity findings and common-effect interpretation; supplement boundary parameterization and Table 1; the practical-homogeneity analysis (Heterogeneity Parameters) and supplement Table 3; the boundary likelihood ratio test in the empirical Results (Between-person heterogeneity).

Comment R1-4

> 4. All nine I^2 values in the empirical illustration exceed 0.94, with several near 0.999. The authors present these as evidence that "the observed dispersion in Stage 1 parameter estimates was dominated by true between-person heterogeneity rather than sampling variability"(p. 25). However, when N = 217 persons each contribute relatively long series (up to 150 occasions after preprocessing), the typical Stage 1 sampling variance $\tilde{v}_p$ is small by construction, inflating I^2. This makes I^2 near unity an almost inevitable artifact of the sample characteristics rather than a substantive finding about affect dynamics. The paper does not discuss this problem; the I^2 metric, developed for traditional meta-analysis where the number of studies is small and sampling variances are large, has different interpretive properties when N is large and Stage 1 estimates are precise. Furthermore, the usefulness of I^2 as a measure of heterogeneity has been questioned in meta-analysis. Most meta-analysts recognize that I^2 is a relative measure of heterogeneity influenced by sample sizes.

Response. We agree that the earlier wording gave I^2 too much substantive weight. The revised text describes it as a measure relative to typical Stage 1 sampling variance, and explicitly notes that more precise individual estimates can produce a large I^2 without greater absolute heterogeneity. We now report the between-person standard deviations in addition to I^2 in the Results section and elaborate on the need for an improved I^2 index for intensive longitudinal models. The General Discussion now treats this dependence on Stage 1 precision as a limitation of I^2 for intensive longitudinal data and outlines alternative summaries of heterogeneity as a future direction.~~The empirical discussion also distinguishes the information in observed reports from the length of the regularized time grid.~~ The revised illustration uses the final analytic sample of 212 participants.

Location in the revision. Manuscript, pp. 12, 26, 28, 42. Definition and interpretation of I-squared; empirical Table 1 (manuscript p. 42); the limitation of I^2 in the General Discussion.

Comment R1-5

> 5. For the bivariate case, τ^2 is 9×9 with 45 free parameters estimated from N persons. As the number of psychological variables k increases, the parameter vector θ_i grows, and the number of free parameters in τ^2 grows hugely. For k = 5 variables (a modest EMA protocol), the Stage 1 parameter vector has dimension ˜55 and τ^2 contains 1,540 free parameters. This is clearly infeasible without strong structural constraints on τ^2. This issue is well-known in MASEM. For instance, the metaSEM package uses diagonal τ^2 as the default option to address this issue. However, the paper does not discuss how MetaVAR scales with k.

Response. We share this concern and have added an explicit account of covariance growth. For the feature vector used here, including the unique innovation-covariance elements gives 45 features at k = 5 and therefore 1,035 distinct heterogeneity components. Although these counts differ from the example in the comment, they support the same conclusion: unrestricted covariance estimation quickly becomes demanding. The Discussion now considers diagonal, block, and theory-selected restrictions, while identifying sparse and low-rank structures as future work. It also explains why parallelizing Stage 1 does not solve the information constraints at Stage 2.

Location in the revision. Manuscript, pp. 32–33. Parameter growth and covariance restrictions.

Comment R1-6

> 6. Only the MetaVAR was used in the empirical illustration. I suggest including the BMLVAR and uncertainty-uncorrected approaches in this illustration. The authors should discuss whether the findings from the simulation studies are also evident in the empirical illustration.

Response. The empirical comparison is much more informative with the uncorrected analysis included. We now present MetaVAR and Uncorr using the same nine-feature Stage 1 estimates. Set-point means are nearly identical, whereas inertia means and several dispersion estimates differ. This is consistent with uncertainty propagation affecting some features more than others, although the empirical data cannot establish which estimator is more accurate.

Following your suggestion, we added DSEM to the empirical comparison. With default priors and with alternative priors scaled to the affect data (not those of the simulation), the fits ran for about 36 hours each but did not converge: the PA-inertia variance had rank-normalized $\widehat{R}$ values of 1.442 and 1.462, and the inertia means also mixed poorly. Because recovery of random-effect covariances can depend on the inverse-Wishart prior when variances are small, we then refitted DSEM with data-informed priors centered on the MetaVAR estimates, following the recommendation to include a data-based prior in such sensitivity analyses (*Schuurman, Grasman, & Hamaker, 2016*). The covariance priors carried the weight of 212 persons, with either the full MetaVAR between-person covariance matrix or only its diagonal as the prior mean. The full version converged ($\widehat{R} \leq 1.03$ for every parameter), and the diagonal version nearly did ($\widehat{R}$ above 1.05 only for the NA-inertia mean and variance).~~Following your suggestion, we attempted to add DSEM to the empirical comparison using both default and alternative priors. Despite approximately 36 hours of computation per fit—about three days of computation across the two fits—neither achieved adequate posterior convergence. We therefore cannot report a reliable numerical comparison with DSEM: the PA-inertia variance has rank-normalized R-hat values of 1.442 and 1.462 and bulk effective sample sizes below eight, and inertia means also show inadequate mixing. The manuscript reports this limitation and the approximately 36-hour runtimes; the supplement gives a brief diagnostic explanation, with full output and traces in the compendium. Thus, we have implemented the MetaVAR–Uncorr comparison and investigated DSEM, but cannot fully answer the requested DSEM comparison without better-converged fits. We have made that limitation explicit rather than interpreting unstable estimates.~~

The four fits answer the question about whether the simulation findings appear in the empirical data, within limits. Set points agreed across all fits and MetaVAR. With weak priors, DSEM estimated higher inertia means and much smaller transition variances than MetaVAR (e.g., .015 under default and .020 under alternative priors, versus .175, for the NA→PA variance). This is consistent with the simulation, where DSEM's recovery of small transition variances depended on the prior. With data-informed priors, the means and variances moved most of the way toward the MetaVAR values, while the transition covariances followed the prior's scale matrix: near MetaVAR under the full version, near zero under the diagonal version. We interpret this as weak identification of heterogeneity in the dynamics, especially its covariances, by the data alone. Because the data-informed priors reuse the same data, their agreement with MetaVAR is partly by construction and is not independent confirmation; the empirical data cannot establish which estimator is more accurate. The manuscript now reports the four fits and their run times, and the supplement gives the prior construction, diagnostics, and a table of all estimates.

Location in the revision. Manuscript, pp. 26, 28, 42; Supplement, pp. 2–3. Empirical comparison and diagnostic limitations; empirical Table 1 (manuscript p. 42), for MetaVAR and Uncorr; the four DSEM fits (Practical and Methodological Considerations) and their run times in the General Discussion; supplement

Empirical Comparison and Table 4, for the DSEM estimates.

I am concerned that you didn't use alternative priors that make things easier for DSEM. So this part sounds weak to me. To me you need to make the IW prior much more restricted (stronger faith, higher df) around the true values, with a base cov matrix that is diagonal. Did you already do that? Can we talk?

Reviewer 2

Comment R2-1

> Simulation study and heterogeneity parameters: As someone familiar with meta analysis in this framework, it is unclear how heterogeneity was quantified here in the data generating mode and evaluation measures. In general the results on heterogeneity parameters in the Monte Carlo study is very hard to read.

Response. The original terminology did not make the target clear enough. We now define heterogeneity in the data-generating process as the between-person covariance of the underlying set points and transition coefficients. The performance section identifies the six population means and 13 heterogeneity components being evaluated, and the Results distinguish mean recovery from variance and covariance recovery. This makes clear that the heterogeneity results concern population variance–covariance parameters, not the estimation of each person's realized random deviation.

Location in the revision. Manuscript, pp. 14, 18, 21. Definition of heterogeneity, recovery targets, and heterogeneity Results.

Comment R2-2

> The paper seems very interested in being able to detect heterogeneity in parameters. While some variances of parameters across individuals are close to zero (i.e., low heterogeneity in the data generating values), it was curious to see that no parameters were explicitly set to have no heterogeneity across people. This would help to assuage concern that homogenous parameters might look heterogeneous due to noise, large sample size, etc. in some modeling frameworks (Hoekstra, Epskamp, and Borsboom, 2023, is a nice motivator of this). While it's great to take inspiration from an empirical example, when the parameters are nearly zero (as in the variance across people) then perhaps it is safe to assume the population-generating parameters is zero, and generate you data accordingly.

Response. The revised simulation now includes an exact-zero transition-heterogeneity condition while retaining heterogeneous set points. This directly examines whether uncertainty in idiographic estimates can appear as between-person variation, the concern raised by *Hoekstra, Epskamp, and Borsboom (2023)*, which we now cite when introducing this condition. ~~Uncorr continued to estimate positive transition variation; MetaVAR reduced that distortion, but its Wald intervals also showed important boundary limitations. In addition, we now report interval exclusion of zero separately from detection of nonzero effects.~~ The supplement includes a separate zero-target figure, and the Methods explain why relative bias is undefined at zero. DSEM requires separate treatment at this boundary. Its variance parameters have strictly positive support, so a posterior credible interval for a variance cannot include zero, even when the true variance is zero. DSEM interval exclusion at exact-zero variance targets is therefore not a test of homogeneity; the manuscript and the supplement figure report it as such rather than as a detection. To address the concern that homogeneous parameters might look heterogeneous, we also report how often each method concluded heterogeneity beyond a small tolerance (a between-person standard deviation of .03) when the true transition variances were zero. Uncorr did so for every estimate and default-prior DSEM for .761 of them, compared with .023 for MetaVAR; averaged over the four

transition variances, MetaVAR's rate was highest, .165, with 200 participants and 50 occasions, consistent with finite-series error in the Stage 1 estimates becoming consequential as the synthesis grows more precise.

The revised Data-Generating Process reads: "This condition examines whether estimation noise in person-specific models can make homogeneous parameters appear heterogeneous (Hoekstra, Epskamp, & Borsboom, 2023)." The Results conclude: "With a tolerance, MetaVAR seldom mistook homogeneous transitions for heterogeneous ones except with many participants and short series, but its intervals were usually too wide to establish homogeneity unless individual series were long."

Location in the revision. Manuscript, pp. 14, 19, 21; Supplement, p. 12. Zero-heterogeneity design and inference; supplement Figure 6 (p. 12); the practical-homogeneity analysis (Heterogeneity Parameters) and supplement Table 3.

Comment R2-3

> Heterogeneity Parameters needs to be defined. It enters into the manuscript in the results section and isn't described in the description of measures. The Performance Measures is a bit vague. It would help to be clear exactly which parameters will be tested for coverage, power, etc. and the inferences made from that. It wasn't clear at first when reading the results why bias in random effects was in the "heterogeneous parameters" section.

Response. We have moved the definition into the Methods, before the results are introduced. "Heterogeneity parameters" now refers explicitly to the 13 freely estimated between-person variances and covariances of the six synthesized features. The performance section states which targets enter each metric: bias and RMSE assess point recovery, coverage assesses interval calibration, and detection assesses sensitivity to nonzero targets. It also distinguishes these population parameters from individual deviations.

Location in the revision. Manuscript, pp. 18–19. Performance Metrics and point-estimate measures.

Comment R2-4

> I took the heterogeneity parameters to be the I^ 2 index (as described on page 11). Yet when we get to the results, it seems that random effects covariances and variances are what is used directly (second sentence of this subsection) with bias and coverage for these estimates reported. It also seems like bias of the random effects was considered under "heterogeneity parameters". This is all very confusing. I saw no reference to I^2 anywhere in the paper after being introduced despite it initially being offered as an assessment of heterogeneity. I had expected bias in I^2 to be explored.

Response. ~~Thank you for identifying this ambiguity.~~ We now distinguish the heterogeneity covariance, which is a simulation recovery target, from I^2, which is used descriptively in the empirical illustration. The framework explicitly states that the simulation does not evaluate recovery of I^2. The empirical Results and Discussion explain its dependence on Stage 1 precision and accompany it with standard deviations in the original parameter units. We also agree with the broader point behind your expectation. I^2 is not an adequate effect size for heterogeneity in intensive longitudinal data: because long series make person-specific estimates precise, it approaches one whether between-person variation is large or modest, as every value above .93 in our illustration shows. The General Discussion now states this limitation, recommends reporting between-person standard

deviations and prediction intervals instead, and outlines heterogeneity indices scaled to the dynamic system as a direction for future work.

The General Discussion now reads: "Long individual series make person-specific estimates precise, and I^2 then approaches one whether between-person variation is substantively large or modest." It continues: "For this reason, we recommend reporting the elements of $\boldsymbol{\tau}^2$ on the scale of each feature, as standard deviations with interval estimates, together with prediction intervals that describe the range of a new person's parameters (Borenstein, 2024; Higgins, Thompson, & Spiegelhalter, 2009; IntHout, Ioannidis, Rovers, & Goeman, 2016)."

Location in the revision. Manuscript, pp. 12, 18, 26, 42. I-squared versus covariance recovery; empirical Table 1 (manuscript p. 42); the limitation of I^2 in the General Discussion.

Comment R2-5

> Relatedly, some parameter values in the A matrix have very low variance across people. Some discussion of the interpretability of this is needed. It was hard to ascertain exactly what was done, but it seems like power to detect effects was taken variance estimates (in addition to random effects). Some variances across people in the A matrix are very small (.001) - if this is not detected as significant, doesn't that make sense?

Response. Yes—low detection of a very small nonzero variance need not be surprising or substantively troubling. We now explain that detection must be read alongside the target's magnitude, bias, RMSE, and coverage. The revised Results describe how detection changes with series length and heterogeneity and explain why higher uncorrected detection can be misleading when coverage is poor. Exact-zero targets are handled separately because inference at a variance boundary poses a different problem.

Location in the revision. Manuscript, pp. 20–22. Interpretation of detection for small nonzero and exact-zero targets.

Comment R2-6

> One benefit of two-stage random effects for VAR models is that you can include all covariates in the VAR model. Anecdotally (and in all tutorials on the method), DSEM seems to have a hard time estimating models with a larger number of variables than 2. It would strengthen the paper to show that MetaVAR can perform well with a larger set of variables.

Response. A larger-variable study would be valuable, but it is beyond the evidence established here. Following the Associate Editor's suggested alternative, we have added a substantive discussion of dimensionality: growth in the parameter vector and covariance matrix, the demands on both stages, and the assumptions introduced by covariance restrictions. The paper now states plainly that the bivariate evaluation does not demonstrate higher-dimensional performance.

Location in the revision. Manuscript, pp. 32–33. Higher-dimensional limitations and future work.

Comment R2-7

> A figure of power results would help. As a general rule if you have more that 10 values in a paragraph you should probably put it in a table or a figure. To compare "power; results, for instance, i had to highlight all uses of that word in the results section and then track the results. This is onerous for the reader, a figure would be much easier (or a table if you want to put all results together).

Response. We now present the power results as figures. Empirical power (detection) for every nonzero target is shown in manuscript Figure 1, by target type, heterogeneity condition, and method, and in Supplement Figure 5, by parameter, sample size, series length, heterogeneity condition, and method. We have made the detection results visible in the figures and reduced the need to track numbers through the prose. Main Figure 1 summarizes detection alongside coverage and zero-target exclusion; the supplement's detection figure shows the individual parameters and conditions, with zero-target exclusion in a separate figure. The Results now explain the effects of sample size, series length, and heterogeneity across the design, including where MetaVAR helps, where the methods are similar, and where DSEM performs better.

Location in the revision. Manuscript, pp. 20, 22, 43; Supplement, pp. 11–12. Detection Results; manuscript Figure 1 (p. 43) and supplement Figures 5–6 (pp. 11–12).

Comment R2-8

> Simulation studies 10-12 had heterogeneous T_i (i.e., some individuals had 50, some had 100, some had 200). I could not find discussion of the impact of this design on outcomes.

Response. We now describe both the allocation and what the unequal-length conditions show. Lengths cycle through 50, 100, and 200, with mean lengths of 115–116.25 depending on participant sample size. At matched N and heterogeneity, MetaVAR and DSEM generally maintained coverage close to their balanced-length counterparts, whereas Uncorr generally deteriorated. MetaVAR's heterogeneity RMSE improved in 80 of 117 comparisons with balanced $T = 100$, versus 10 for Uncorr. We also state the important qualification: the unequal-length conditions contain more observations on average, so this comparison does not isolate unequal precision at a fixed information budget.

Location in the revision. Manuscript, pp. 16, 22. Unequal-length allocation and condition effects.

Comment R2-9

> Page 7: when saying "In Stage 1, each individual .... Is fit with an idiographic dynamic model" perhaps specify that this is done outside of the random effects model in SEM. As in, it can be estimated with any VAR package of their choosing.

Response. We have clarified that the idiographic models are fitted separately, outside the Stage 2 random-effects SEM. Stage 2 can accept estimates from another suitable estimator, provided it supplies compatible parameter definitions and their aligned joint sampling covariance. This requirement is important: a list of coefficients and standard errors alone does not preserve the cross-parameter uncertainty that MetaVAR is designed to carry forward.

Location in the revision. Manuscript, pp. 7–8. Separate Stage 1 estimation and required outputs.

Comment R2-10

> When describing the empirical analysis you describe that you use the A matrix and covariance matrix of residuals as inputs to the meta-analytic framework. What about the sampling covariance matrix that quantifies estimation uncertainty?

Response. You are right: the earlier description omitted a central input. The empirical section now states that each participant contributes both the nine-feature estimate vector and its full 9-by-9 sampling covariance matrix. This is distinct from the innovation covariance within the VAR. We checked the transfer against the saved Stage 1 fits, and the supplement briefly records the numerical conditioning applied to the sampling covariances. The revised account makes explicit where first-stage uncertainty enters the synthesis.

Location in the revision. Manuscript, pp. 25–26; Supplement, p. 2. Empirical Stage 2 inputs and sampling-covariance conditioning.

Comment R2-11

> Sometimes BMLVAR is used, sometimes DSEM, for what seems to be the same one. DSEM is the more popular term, so it makes sense to use that throughout.

Response. We have standardized the manuscript and figure terminology to DSEM and defined it when introduced. This avoids suggesting that BMLVAR and DSEM denote two different comparison methods.

Location in the revision. Manuscript, pp. 2, 17. DSEM terminology introduced in the Abstract and specified in Methods (manuscript p. 17).

Comment R2-12

> Some rationale for using fitVARMxID is needed. Why use this when other approaches (such as MIIV-2SLS) have been shown to provide maximum-likelihood equivalent estimates in the SEM framework? Why not use Kalman Filter estimation (i.e., the vars package)? Is MetaVAR flexible and the user can choose? (even though simulation results are grounded to this package)?

Response. We have added the rationale for using the fitVARMxID R package. It fits each person's centered VAR as a state-space model in OpenMx and evaluates the likelihood with the Kalman filter (Hunter, 2018). The Kalman filter accommodates unobserved occasions within the likelihood, which matters in the empirical illustration, where about 83% of the hourly grid entries were unobserved. The same fit returns the full joint sampling covariance of all estimated features, which Stage 2 requires. Your question about alternatives prompted a closer comparison. The `vars` package estimates each VAR equation by ordinary least squares rather than with the Kalman filter, and it does not accept series with missing values. MIIV-2SLS estimates each equation separately with model-implied instrumental variables; it is consistent and robust to some forms of structural misspecification. In a lagged time series, however, a missing occasion removes both the row in which it is the outcome and the row in which it is a predictor, so listwise handling would discard most of the empirical data. We therefore used a full-information likelihood with the Kalman filter for Stage 1.~~Its centered, likelihood-based VAR workflow provides the parameter vector and full joint sampling covariance needed for Stage 2, including missing-data handling.~~ The same passage explains that MetaVAR is not conceptually tied to this package:

another estimator can be used if it supplies compatible outputs. We limit the simulation conclusions to the implementation evaluated here and do not assume that alternative estimators or packages provide equivalent estimates or uncertainty information.

Location in the revision. Manuscript, p. 8. fitVARMxID R package rationale and flexibility of Stage 1.

Comment R2-13

> In the Performance Measures section it would be helpful to say how each one will be used to expectations that are clearly laid out in the preceding subsection.

Response. The performance section now connects each measure to the preceding expectations: point recovery is assessed with bias and RMSE, uncertainty calibration with coverage, and sensitivity to nonzero effects with detection. The Results follow these distinctions. They show why propagating uncertainty helps recover transition heterogeneity relative to Uncorr, while also recognizing DSEM's stronger recovery of transition means and the limits of Wald inference at zero.

Location in the revision. Manuscript, pp. 18, 20. Comparative expectations, performance measures, and Results.

Comment R2-14

> Tau-squared is reported for the empirical study - perhaps add to the simulation study. What are the true values in the data generating parameters? The obtained values?

Response. We now identify the simulation heterogeneity targets explicitly and distinguish the displayed generating covariance from the calibrated recovery targets after stability rejection. The Results explain recovery of the transition variances and covariances, and the supplement presents the full condition-level bias, RMSE, and coverage figures. For direct inspection of true and obtained values, the `results` data set in the research compendium (`data/results.rda`, documented on the compendium website) contains the calibrated target values, mean estimates, bias, RMSE, coverage, power, and their Monte Carlo standard errors~~the compendium file validated-performance.csv contains the target values, mean estimates, and Monte Carlo errors~~ for every parameter, condition, and method. ~~Its location is .setup/notes/manuscript/MBR-26-107-R1/revision-execution/evidence/validated-performance.csv.~~ We provide this complete numerical record electronically rather than reproduce hundreds of rows in the supplement, so readers can inspect individual targets without obscuring the main patterns.

Location in the revision. Manuscript, pp. 14, 21; Supplement, pp. 4, 8–10. Generating covariance and recovery Results; supplement figure guide and Figures 2–4 (pp. 8–10).

Comment R2-15

> I may have missed it, but was there a discussion of how the random effect estimates may differ from the individual-level estimates? That is, are those thought to be improved upon over solely individual-level analysis

Response. We have added this distinction to the empirical Discussion. MetaVAR's population means and heterogeneity covariances summarize the distribution of person-specific parameters; they do not replace the unpooled Stage 1 estimates. We therefore do not claim improved person-level estimation through shrinkage. Establishing such an improvement would require a separate evaluation of individual predictions.

Location in the revision. Manuscript, p. 29. Population synthesis versus person-level estimation.

Comment R2-16

> Is Stage 1 in the Uncertainty-Uncorrected Two-Stage Benchmark identical to Stage 1 in the MetaVAR, with the difference being that uncertainty isn't considered in UUTSB? Please clarify.

Response. Yes. The revised Methods now state that MetaVAR and Uncorr use the same saved Stage 1 fits, the same six point estimates, and the same block restrictions in the simulation. The difference is that MetaVAR carries their sampling covariance into Stage 2. The empirical comparison likewise uses identical nine-feature point estimates. This makes the comparison one of uncertainty propagation as a whole; it does not isolate the contribution of the off-diagonal sampling covariances.

Location in the revision. Manuscript, pp. 18, 26, 42; Supplement, p. 2. Common Stage 1 inputs; empirical Table 1 (manuscript p. 42).